Effect of Team, Generation, and Track on Pit Stop Time in Formula 1

STAT 301-01: Analysis of Variance and Multivariate Analysis

Authors: Abduselam Mohammed
Owen Larson

Professor: Wei Wei

Term: Spring 2026

Institution: Metropolitan State University

St. Paul, MN

Introduction

This project examines Formula 1 pit stop data to study how pit stop time changes across teams and across two different eras of the sport. Pit stop time is a useful performance measure because it reflects both technical efficiency and planning as well as pit team execution, and it has also been affected by changes in rules, strategy, and technology over time. In this analysis, pit stop time is the response variable, while team, recorded as Car, and generation are the main explanatory variables, Country (track location), recorded as Race, was included in an expanded final model to account for track variation. Generation separates the data into two eras, 1994–2005 and 2006–2010, to capture broad differences in Formula 1 conditions across them.

Before the analysis, the data were cleaned by converting pit stop time to numeric format. Rows with invalid or zero duration pit stop times were removed before modeling. In addition to the ANOVA, a correlation scan was used to examine relationships among the numeric variables in the dataset. The most meaningful result from that scan was a moderate negative correlation between year and pit stop time, $r = -0.345$, suggesting that pit stops generally became faster over time. This supports the broader question of whether pit stop performance differs not only by team, but also by era, and whether team differences remain consistent across generations.

Research Questions

1. Do mean pit stop times differ across teams and across generations of Formula 1 cars?

2. Does the relationship between team and pit stop time change across generations, and how does pit stop time trend over the years?

3. Does the particular racetrack appear to contribute additional variation in pit stop time beyond team and generation?

Variables Used in the Analysis

Variable	Type	Meaning
Time	Quantitative	Pit stop duration for a single stop
Car	Categorical	Team or constructor name in the dataset
Generation	Categorical	Two eras: 1994–2005 and 2006–2010
Year	Quantitative	Season year, used for time trend and correlation scan
Race	Categorical	Country (track) where the race was held.

Hypotheses

For the main effect of team, H0: the mean pit stop time is the same across all teams.

Ha: at least one team has a different mean pit stop time.

For the main effect of generation, H0: the mean pit stop time is the same across generations.

Ha: mean pit stop time differs between generations.

For the main effect of race, H0: the mean pit stop time is the same across races.

Ha: at least one race has a different mean pit stop time.

For the interaction between team and generation, H0: there is no interaction between team and generation on pit stop time. Ha: there is an interaction between team and generation on pit stop time.

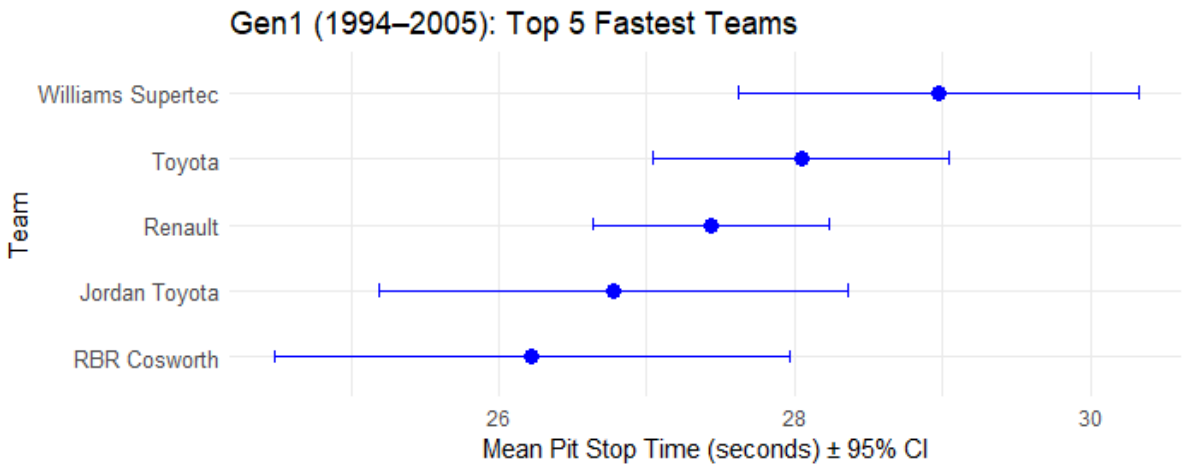
Methods

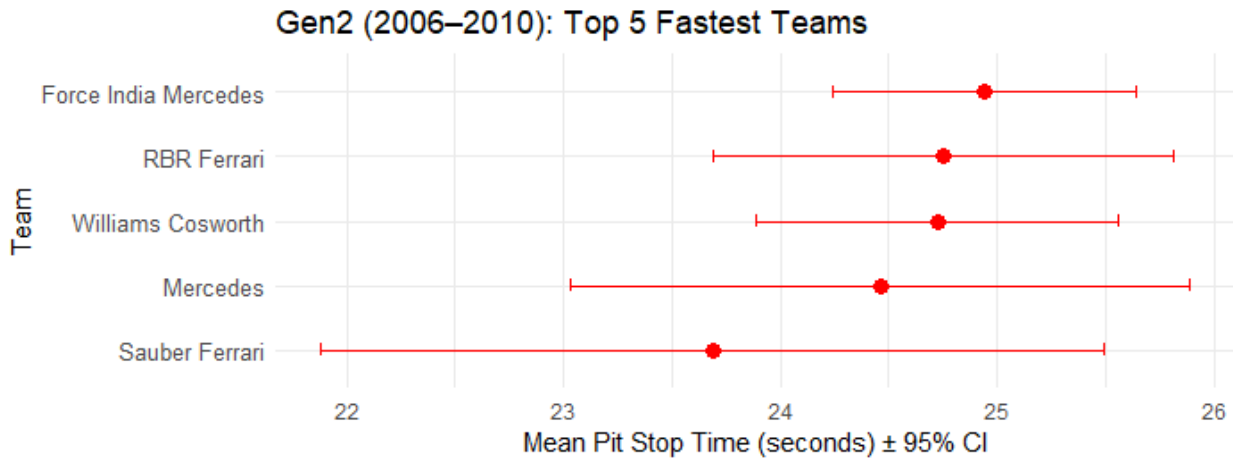
The data were analyzed using ANOVA methods designed to compare pit stop times across teams, generations and race contexts. A factorial ANOVA model with an unbalanced design was used for the main analysis, because the response variable was quantitative and the explanatory variables: team, race, and generation, were categorical. Since the numbers of observations were not equal across combinations of team, generation, and race, Type II sums of squares were used. A correlation analysis was also performed on the main numeric variables in the dataset as a supplementary descriptive check. Before fitting the final model, the data were cleaned by removing rows with missing team labels and reconstructing one missing pit stop duration. The distribution of pit stop time was then examined for extreme values. Because the goal of the project was to study pit stop performance rather than unusually long garage related events, pit stop durations above the chosen upper threshold (127s, 99th percentile) were excluded from the main analysis. A log transformation of pit stop time was then used for the final two-way ANOVA, residual diagnostics improved under the transformed model.

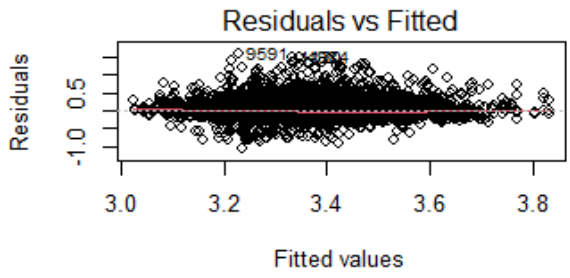
The main statistical model for the final analysis was $\text{logTime} = \mu + \text{Team} + \text{Generation} + \text{Race} + \text{Team} \times \text{Generation} + \varepsilon$. The final model showed a significant main effect of team: $p < 0.001$, a significant main effect of generation: $p < 0.001$, and a significant main effect of race: $p < 0.001$. The interaction between team and generation: $p = 0.0002789$ remained significant. These results indicate that pit stop performance differed across teams, differed across generations, varied by race, and that the team differences were not constant across the two eras. Because the interaction was significant, the data were then separated by generation and one-way ANOVAs were used to examine team differences within each era. In Gen 1 and 2, team had a significant effect on pit stop time: $p < 0.001$ for both. Tukey post-hoc comparisons were then used to identify which team pairs differed significantly within each generation. The largest significant differences were observed between RBR Cosworth vs Pacific Ilmor for Gen 1 and Sauber Ferrari vs Force India Ferrari for Gen 2.

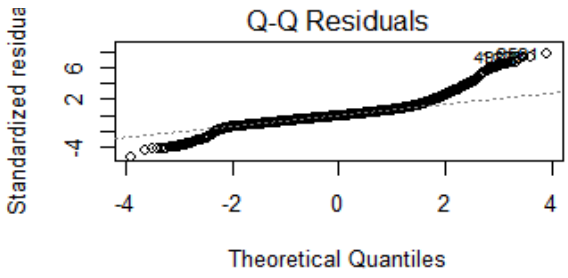
Summary of Main Statistical Results

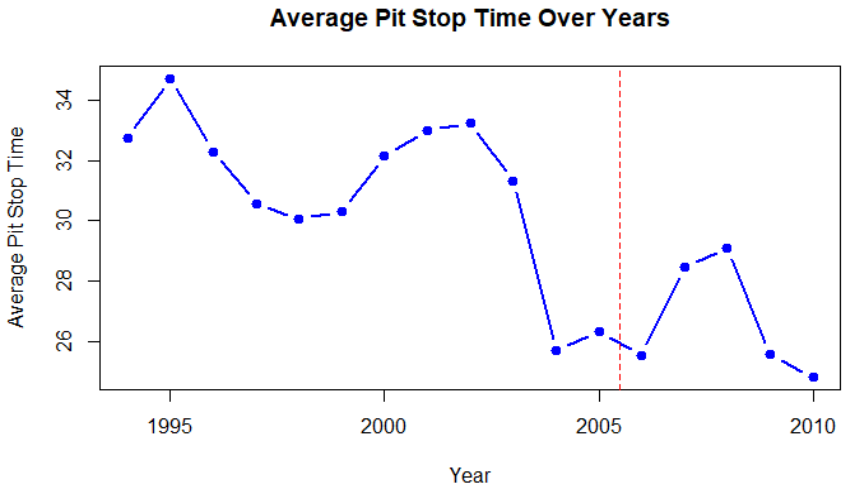
Analysis(logTime)	Effect	df	F	p-value
Two-way ANOVA	Team	70, 9716	15.7924	< 0.001
Two-way ANOVA	Generation	1, 9716	132.9651	< 0.001
Two-way ANOVA	Race	26, 9716	45.4084	< 0.001
Two-way ANOVA	Team × Generation	3, 9716	6.3253	0.0002789
Gen1 one-way ANOVA	Team	50, 6597	10.95	< 0.001
Gen2 one-way ANOVA	Team	23, 3145	6.839	< 0.001

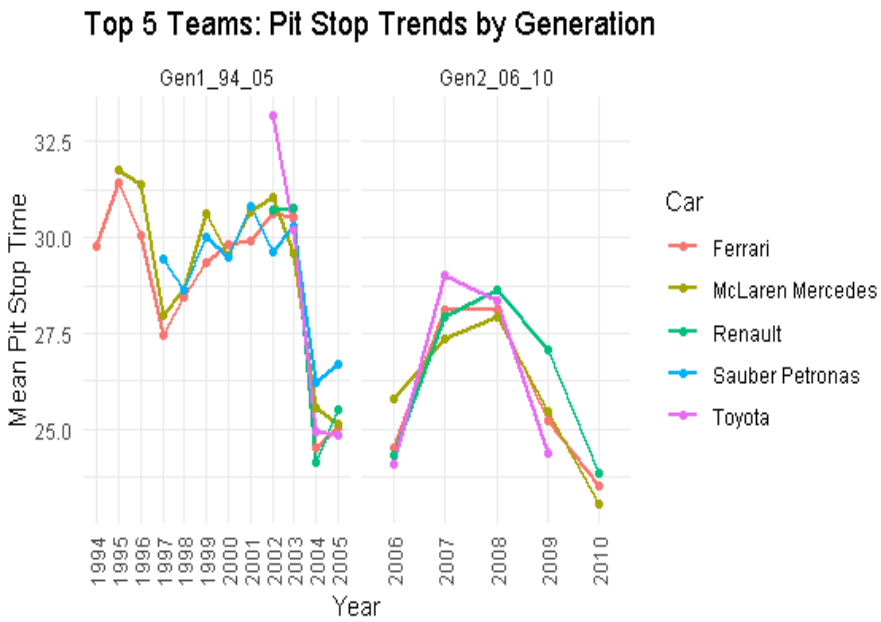












Discussion

The results show that pit stop performance is influenced by both team and generation, and that these effects are not independent. The significant interaction between team and generation indicates that team performance changes across eras, meaning that some teams adapt more effectively to regulatory and technological changes than others.

Race was also significant in the final model, which suggests that pit stop times were influenced not only by team and era, but also by event specific conditions. This likely reflects differences in pit lane layout, track context, and race strategy across racetracks. Even after accounting for different tracks, the effects of team, generation, and the interaction between team and generation remained significant, which strengthens the main conclusions of the analysis.

The observed decrease in pit stop times around 2004–05 can likely be explained by changes in Formula 1 regulations. In 2005, tire change restrictions required drivers to complete races on a single set of tires when possible, reducing the complexity of pit stops; resulting in shorter times. Additionally, this period reflects a peak in pit crew efficiency, with teams optimizing equipment and coordination.

When tire changes were reintroduced in 2006, pit stop times began to rise again relative to the unusually low values seen in 2005. This increase became more noticeable in 2007, when teams were also required to use both dry tire compounds during a race, making tire strategy a more consistent part of pit stop execution. In 2010, refueling was no longer allowed during the race, which coincided with another clear drop in mean pit stop time. Together, these rule changes suggest that shifts in pit stop duration were tied not only to team performance, but also to changes in what a pit stop was required to include.

The Tukey analysis further reveals that certain teams exhibit significantly better performance than others, and that the magnitude of these differences varies between generations. This suggests that while overall improvements in pit stop efficiency occur over time, competitive advantages between teams are quite dynamic rather than fixed.

Conclusion

At a significance level of 0.05, the results indicate that pit stop times differ significantly across teams, generations, and tracks. There was also a significant interaction between team and generation. This demonstrates that team performance is not consistent across eras and is influenced not only by changes in regulations and technology, but also by track context. Despite short term fluctuations associated with the changes in fueling and tire regulations, the overall pattern shows that average pit stop times decreased across the years present in this study. This suggests that long term improvements in pit crew execution, equipment, and team strategy outweighed the temporary effects of rule changes.

Together, these findings show that pit stop performance in Formula 1 is influenced by team, era, and race context at the same time. Among the effects of the final model, generation showed the strongest statistical effect, suggesting broad changes in regulations and technology had the largest overall impact on pit stop duration. Race, or track location, was the next strongest effect, showing that specific track conditions also mattered substantially, while team remained significant as well. The significant interaction between team and generation shows that team differences were not fixed across eras. This means that pit stop duration reflected not only differences in execution between teams, but also broader changes in the sport and the conditions under which each race was run.

Limitations

There are several limitations to this analysis. The dataset is unbalanced, with unequal representation of teams across years and generations. Although race was included in the final model, the analysis still cannot isolate which specific race characteristics, such as pit lane length, pit lane speed limits, typical weather, or strategy patterns, were responsible for the observed race effect. Additionally, team names may change over time due to sponsorship or organizational changes, which may affect continuity in the data. Other factors such as driver behavior, race conditions, and weather were not included in the analysis and may also influence pit stop time.

Despite these limitations, the results provide meaningful insight into how pit stop performance evolves across teams and regulatory eras.

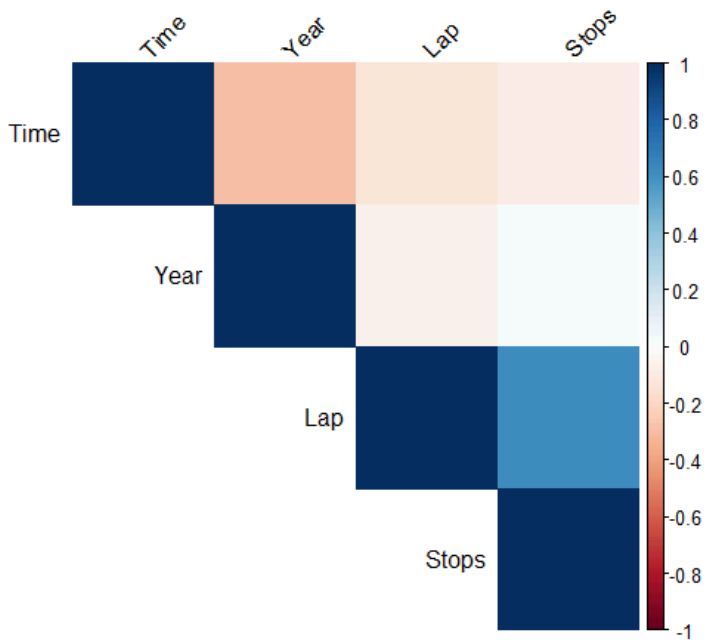
Github links:

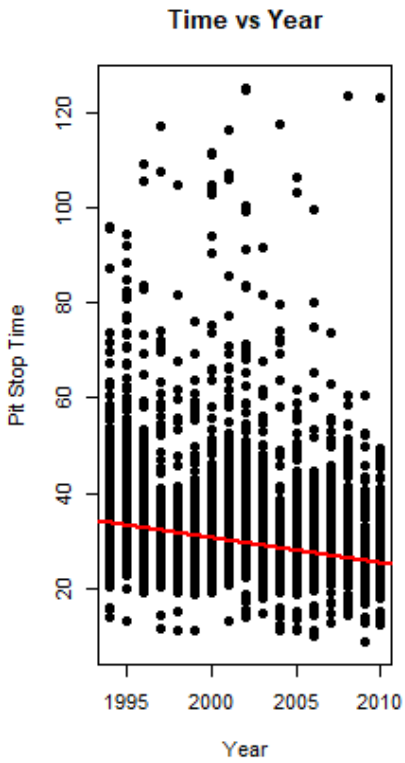
[Abduselam2020/Effect-of-Team-and-Year-on-Pit-Stop-Time-in-Formula-1]

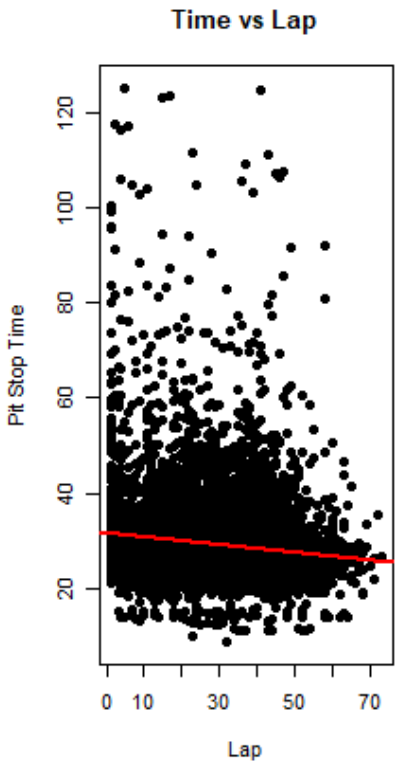
Sources:

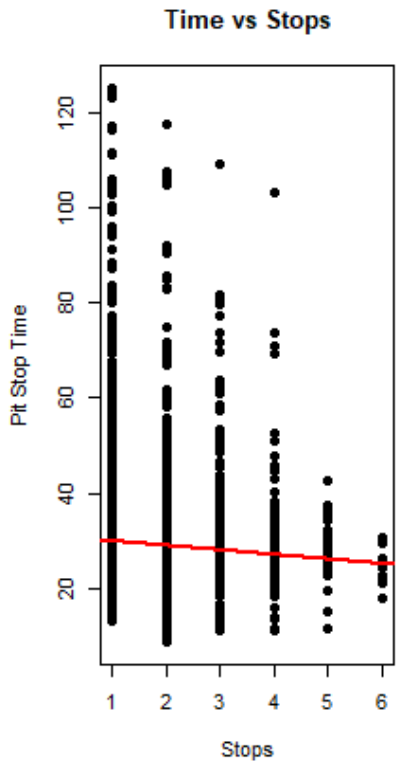
https://en.wikipedia.org/wiki/Formula_One_regulations

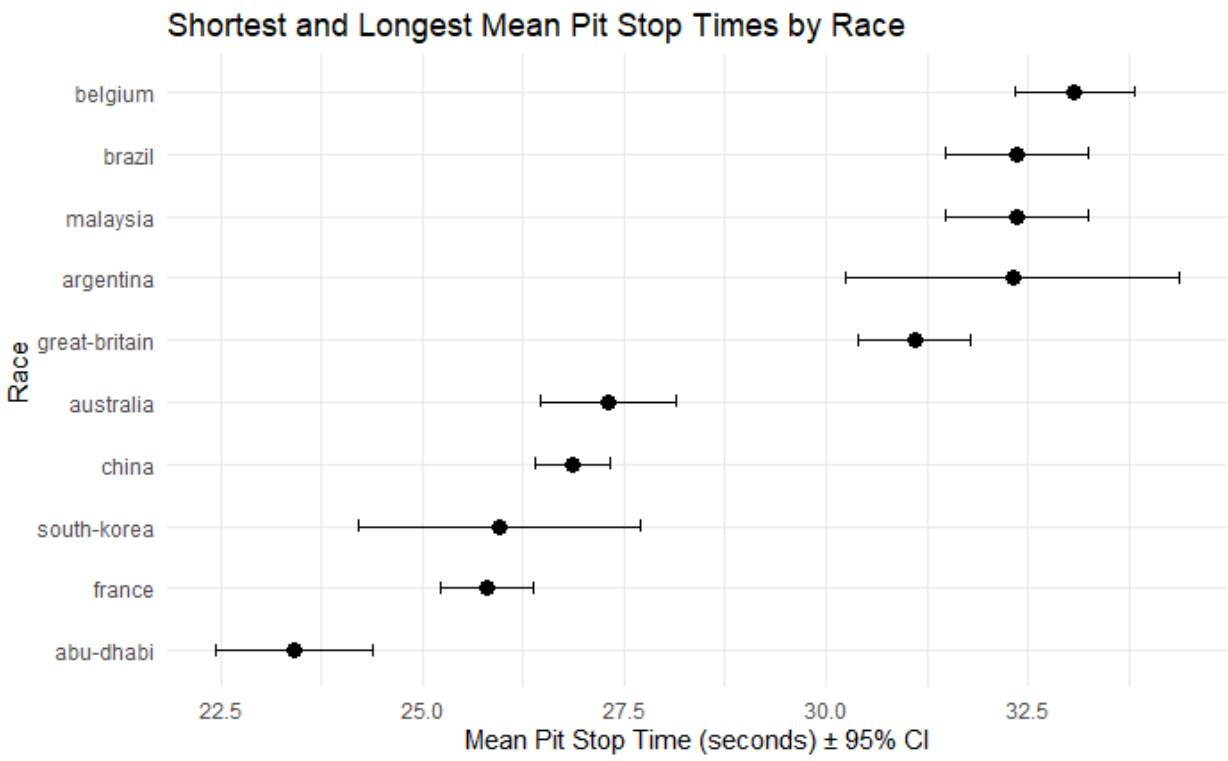
Appendix:

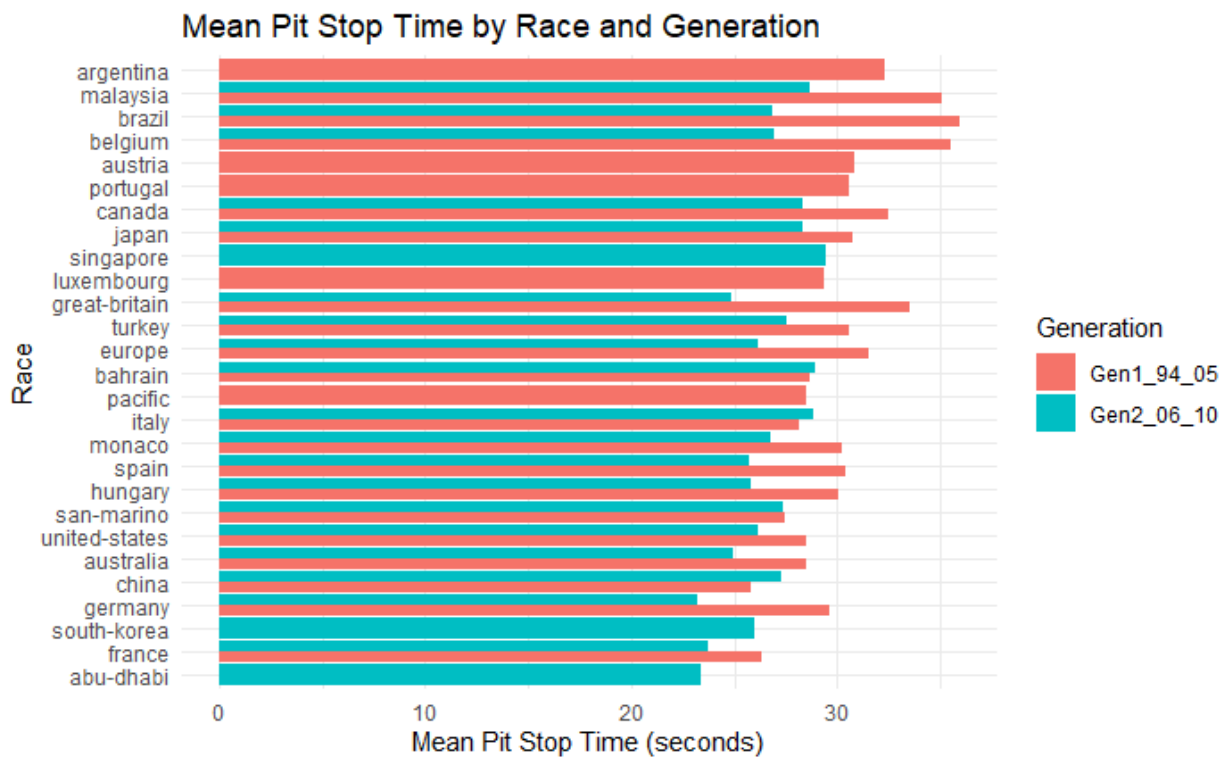









Shortest and Longest Mean Pit Stop Times by Race
belgium
brazil
malaysia
argentina
great-britain
australia
china
south-korea
france
abu-dhabi
Race
22.5
25.0
27.5
30.0
32.5
Mean Pit Stop Time (seconds) ± 95% CI


Mean Pit Stop Time by Race and Generation
argentina
malaysia
brazil
belgium
austria
portugal
canada
japan
singapore
luxembourg
great-britain
turkey
europe
bahrain
pacific
italy
monaco
spain
hungary
san-marino
united-states
australia
china
germany
south-korea
france
abu-dhabi
Race
0
10
20
30
Mean Pit Stop Time (seconds)
Generation
Gen1_94_05
Gen2_06_10